

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -1st

Course: B.Tech. Semester: 3rd Session: 2025-26
Subject: Computer Organization & Architecture
Max. Marks: 20

Section: A, B, C, D, I, J
Subject Code: BCS-302
Time: 1 hr 15 min.

Q.No. :	1	2	3	4	5	6
CO No.:	<u>1</u>	<u>2</u>	<u>1</u>	<u>2</u>	<u>1</u>	<u>1</u>
Bloom's Level	<u>2</u>	<u>3</u>	<u>2</u>	<u>3</u>	<u>3</u>	<u>2</u>

Note:

1. This paper contains two section A and B.
2. All sections are compulsory.

faiz

SECTION- A

- Q1. Discuss the role of ALU in a processor? 2 marks
Q2. Perform single and double precision IEEE standard floating-point number representation for $(-1460.125)_{10}$. 3 marks
Q3. Explain Bus Arbitration and its types. 3 marks

SECTION- B

- Q4. Show the multiplication process step by step using Booth's Algorithm for $(-19) * (+20)$.

OR

Divide **39 by 17** using Binary Division Algorithm also draw the flowchart of Division Algorithm. 4 marks

- Q5. Convert the following numerical arithmetic expression into reverse polish notation and show the stack operations for evaluating the numerical result.

$$(3+4) [10(2+6) + 8]$$

Write the PUSH and POP operation of stack. 4 marks

- Q6. Explain General Register Organisation with proper diagram. Also define control word and write control word for following microoperation.

- a. $R5 \leftarrow 0$
- b. $\text{Output} \leftarrow \text{Input}$

4 marks

ALL THE BEST

MORADABAD INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING (AI&ML)
SESSIONAL TEST -1st

Section: D

Course: B.Tech. Semester: 3rd Session: 2024-25
 Subject: Computer Organization & Architecture
 Subject Code: BCS-302
 Max. Marks: 20

Time: 1 hr.

Q.No. :	1	2	3	4	5	6
CO No. :	2	1	1	2	1	1
Bloom's Level	3	2	2	3	3	3

Note:

1. This paper contains three sections (A), (B) and (C)
2. All sections are compulsory.

SECTION - A

Q1. Perform arithmetic shift right, logical shift right and circular shift right consecutively on 11000101.

2 marks

SECTION - B

Q2. Explain Bus Arbitration and its types.

3 marks

Q3. Explain the following addressing modes with example:

- a) Direct addressing mode b) Immediate addressing mode c) Relative addressing mode

3 marks

SECTION - C

Q4. Show the multiplication process step by step using Booth's Algorithm for $(-12) \times (+16)$.

4 marks

Q5. Implement the following arithmetic statement using Assembly Language for one-address and zero-address instructions-

$$X = (A + B \oplus C) / (D - E).$$

4 marks

Q6. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is conducted with multiplexers.

4 marks



- a. How many selection inputs are there in each multiplexer?
- b. What size of multiplexers is needed?
- c. How many multiplexers are there in the bus? 16
- d. What size of decoder is needed? 8

4 x 16.

The Family!

The newly-appointed chief secretary of Jharkhand Alka Tiwari is the third chief secretary in her family. Her husband Devendra Tiwari was also a chief secretary in Jharkhand and is currently the state election commissioner after his retirement. Rajendra Kumar Tiwari, elder brother of Devendra Tiwari, was the chief secretary of UP. Dhirendra Tiwari, younger brother of Devendra, is currently serving as additional chief secretary in Punjab. The Tiwari family hails from a village in UP and is probably one of the few families to have three chief secretaries in a state.